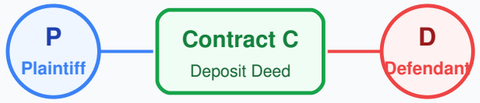
SynapseLex Neuro-Symbolic Judicial Adjudication Architecture

Bochner Continuous Dual-Clock Attention + OWL 2 DL TBox Arbiter + Defeasible Dung Head

STAGE 1 Dynamic Legal Graph

Multi-Relational Extraction: G(t) = (V, E, T, S)

Bipartite Dispute Subgraph



Consideration: 500M VND | **Form:** Unnotarized

Performance Ratio: 350M/500M = 70%

-> Triggers Statutory Cure (Art. 129.2 >= 2/3)

Chronological Timeline Events

- **t_0: Contract Execution**
  Valid consent (Art. 117 BLDS 2015)
- **t_1: Refusal & Default**
  Defendant repudiates transfer deed
- **t_2: Tolling Suspension (Art. 156)**
  COVID-19 Court Closure (dt/dtau = 0)
- **t_3: Lawsuit Filing at Trial Court**
  Elapsed latency: Delta tau = 1,254 days
  Over 3-year bar (Art. 429 > 1,095d)

G(t)

STAGE 2 Axiom-TGAT Attention

Continuous Bochner Harmonic Deontic Continuum

Continuous Bochner Fourier Basis

$$\Phi(\Delta\tau) = \sqrt{2/d}\,[\cos(\omega_k \Delta\tau), \sin(\omega_k \Delta\tau)]^T$$

Dual-Clock Procedural Time Mapping:

- Calendar Time: $t \in \mathbb{R}^+$ (Astronomical days)
- Procedural Clock: $\tau(t) \in \mathbb{R}^+$ (Statutory timeline)

Art. 156 Tolling: dtau/dt = 0 (COVID freeze)

Art. 157 Restart: tau <- 0 (Debt acknowledgment)

Multi-Head Continuous Temporal Attention

$$\alpha_{uv}(t) = \mathrm{Softmax}\left(q_u^T [k_v \,\|\, \Phi] / \sqrt{d_k} \right)$$

Inductive Temporal Attention Weights

$$h_u(t) = \sum_v \alpha_{uv}(t) \left(W_v [z_v(t) \,\|\, \Phi] \right)$$

2-Hop Continuous Neighborhood Aggregation

Dynamic Representation Output:

$$z_{uv}(t) = \mathrm{MLP}([h_u(t) \,\|\, h_v(t) \,\|\, e_{uv}(t)])$$

Latency: **8.15 ms/case** on NVIDIA A30 GPU

z_uv

STAGE 3 Ontological Arbiter

OWL 2 DL TBox Loss + Markovian State Gating

Vietnamese Civil Law Ontology (VCLO)

r_1: Lapsed <= Extinguished (Art. 429)

r_2: TimeBarred <= ~EnforceableRelief

r_3: Perf >= 2/3 <= CuredDefect (Art. 129.2)

r_4: UsuriousRate <= PartialVoid (Art. 468)

HermiT Reasoner: 100% Satisfiable (0 contradictions)

Normative Hierarchy Priority (NH-DA):

Lex Superior: Const. > BLDS 2015 > Decrees

Lex Specialis: Land Law 2024 > BLDS 2015

Continuous Differentiable DL Loss

$$L_{onto} = \sum_k \max(0, \psi_k^{\mathrm{smooth}}(\theta))$$

Lukasiewicz t-norm relaxation (No gradient void)

$$L_{total} = L_{task} + \lambda_s L_{state} + \lambda_{onto} L_{onto}$$

Joint Multi-Objective Optimization (lambda_onto = 1.0)

Logical Consistency Rate (LCR):

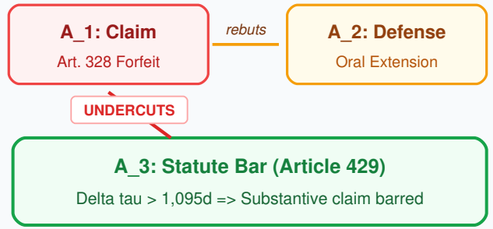
99.87% LCR (TOVR = 0.13%)

M_def

STAGE 4 Defeasible Adjudication

Phan Minh Dung Abstract Argumentation Framework

Defeasible Argument Graph AF = (Args, Att)



Grounded Extension: GE(AF) = [A_3]

Knaster-Tarski Adjudication & SOTA

$$F_{AF}(\emptyset) = [A_3] \Rightarrow GE(AF) = [A_3]$$

Least Fixed Point Grounded Extension

$$z_{masked} = z + \log(M_{defeasible})$$

Strict 100% Soundness Guarantee

AUDITED VERDICT: DISMISSED (CODE 2)

Statute-Barred Claim Extinguished

90.17% Hits@1 | 100% Statute Rec.